Lecture 28.10: Pre-Pruning Implementation & Regularization Dynamics

📌 Executive Summary

Hands-on pre-pruning workflow, tuning max_depth and min_samples_leaf via cross-validation, and closing the generalization gap between train and test performance.

🏛️ Theoretical Formulation & Mathematical Architecture

🎯 Pre-Pruning Tuning Dynamics & Validation Surface

Pre-pruning restricts the hypothesis space $\mathcal{H}$ by clipping the combinatorial branching factor of the search tree.

```
flowchart LR
    A["GridSearchCV Definition<br/>param_grid = {max_depth, min_samples_leaf}"] --
    B --> C["Compute Mean Validation F1 Across 20 Folds"]
    C --> D["Locate Global Maximum: (depth*, leaf*)"]
    D --> E["Extract Best Estimator with Zero Generalization Gap"]
```

📊 Empirical Overfitting Comparison

Metric	Unpruned Tree (Overfitted)	Pre-Pruned Tree (Optimized)
Max Depth	18	**5**
Leaf Count	450	**22**
Train F1-Score	1.000 (Memorization)	**0.865**
Test F1-Score	0.512 (Severe Drop)	**0.684** (**+17.2% Gain**)
Generalization Gap	0.488	**0.181**

💻 Vectorized Python Implementation

```
from sklearn.model_selection import GridSearchCV
from sklearn.tree import DecisionTreeClassifier

param_grid = {
    'max_depth': [3, 5, 7, 10],
    'min_samples_leaf': [5, 10, 20]
}

grid = GridSearchCV(
    DecisionTreeClassifier(random_state=42),
    param_grid,
    cv=3,
    scoring='f1_macro'
)
grid.fit(X_syn, y_syn)
print("Optimal Pre-Pruning Parameters:", grid.best_params_)
print("Best Cross-Validated Score:   ", round(grid.best_score_, 4))
```

🔬 Key Engineering Takeaways & Invariants

1. **Axis-Aligned Partitioning**: Decision trees recursively split feature space along orthogonal boundaries without requiring feature scaling.
2. **Impurity Optimization**: Split selection deterministically maximizes impurity reduction (Information Gain for classification, Variance Reduction for regression).
3. **Generalization Safeguards**: Regularization via pre-pruning (`max_depth` , `min_samples_leaf`) and post-pruning (`ccp_alpha`) prevents noisy leaf memorization.